

National Institute of Technology Raipur

Department of Electronics and Communication Engineering

SEMESTER PROJECT (PROJECT-I) PRESENTATION

16-bit ALU: RTL to GDSII Implementation and Optimization on SKY130A

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Introduction

Context

The ALU is among the most exercised datapath blocks, so its speed, area and power shape processor performance and cost.

Platform

One 16-bit ALU taken from RTL to GDSII with OpenLane 2 on the SkyWater SKY130A PDK (sky130_fd_sc_hd).

Approach

The same function implemented three times (delay, area-power, hybrid) and compared with each other and with published work.

RTL-to-GDSII flow



Objectives & Motivation

Objectives

- Take a registered 16-bit ALU from RTL to DRC- and LVS-clean GDSII
- Optimize three ways: delay, area-power and a balanced hybrid
- Show the baseline, every design step and the numbers behind it
- Show how published papers guided each change

Motivation

- Speed, area and power pull in different directions in the ALU
- Textbook estimates rarely survive a real cell library and P&R flow
- Open-source tools let a student team measure every choice on a real PDK

Literature Survey: ALU Papers

Paper	Scope and reported result	Role in our design
[1] Ravindran & Lourde, ICTRC 2015	4-bit adders: CLA 1.49 ns / 1.50 mW vs RCA 1.55 ns / 0.76 mW. 16-bit ALU: CMOS 2.84 ns vs mixed logic 2.75 ns (PDP −1.95%).	CLA buys speed at about twice the power: basis for A's CLA and B's compact RCA.
[3] Uniyal & Niranjan, ICCCA 2017	8-function 16-bit ALU, variable-stage adder + pass-transistor mux, 90 nm: −23.48% delay, −2.76% power.	Closest analogy: adder and operation select designed together (C), decode before the flops (A).
[2] Santhanam & Prasad, ICICACS 2025	Verilog 16-bit vs 32-bit ALU on Spartan-6: 10.365 ns / 128 mW vs 17.586 ns / 237 mW.	FPGA reference for width scaling; not comparable with SKY130 area or power.
[11] Mallikarjuna, PrimeMOS tutorial	4-bit ALU RTL-to-GDSII on SKY130 with OpenLane; not peer reviewed, no post-route benchmark.	Flow reference only (synthesis to GDSII steps).

Literature Survey: Adder Papers

Paper	Scope and reported result	Role in our design
[7] Nagendra, Irwin & Owens, TCAS-II 1996	HSPICE framework across RCA, carry-skip, carry-select and CLA; area, delay and power are coupled trade-offs.	Why area-power (B) and delay (A) are separate objectives.
[6] Kim & Kim, ISCAS 2001	64-bit carry-select with one RCA + add-one circuit: 38% fewer transistors than dual-RCA, 3.45 ns at 2.5 V, 0.25 µm.	Idea behind C's carry-increment upper half and B's shared hardware.
[10] Ramkumar & Kittur, TVLSI 2012	Modified SQRT-CSLA, 8 to 64 bits: less area and power than the regular one for a slight delay increase, 0.18 µm.	Supports PPA trade-off: Run 20's redundant-gating removal and C's split adder.
[9] Saini, Akhter & Chauhan, VLSI Design 2016	RCA, CLA, carry-select, SQRT-CSLA and common-Boolean adders, 8 to 64 bits, 32/28 nm standard cells.	Adder architecture comparison behind A's CLA study; technology differs.

Literature Survey: Shifter & Power Papers

Paper	Scope and reported result	Role in our design
[4] Das & Khatri, TCAS-II 2008	Timing-driven merging of barrel-shifter stages, 28 combinations: average 10.19% faster for 4.11% more area; up to 13.84% faster for 7.29% more area.	Strong timing reference for A's two-level one-hot shifter (structure not identical).
[5] Chon et al., ISOCC 2022	32-bit bidirectional shifter, 28 nm dynamic-logic MUX: 36% power, 19% performance, 37% area gain vs domino logic.	Shows shifter architecture matters; dynamic logic differs from our static cells.
[8] Banerjee et al., ICCD 2005	Operand isolation for low-power datapath synthesis; warns that isolation logic itself costs area and timing.	Motivation for B's operand gating; our Runs 17 and 25 agree with the warning.

Papers use 90, 180, 250 and 28 nm CMOS or FPGA, different tools and circuit styles: their values are context, not benchmarks for our numbers.

Paper Influence: Design A (Delay)

Change in our design	Paper	Paper expects	We measured	Conclusion
CLA adder: grouped lookahead replaces the ripple chain	[1] Ravindran & Lourde	CLA slightly faster than RCA (1.49 vs 1.55 ns) at about twice the power	Isolated adder 8.50 to 7.16 ns; Run 8 to 9: 10.32 to 10.01 ns for +10% area; power rose	Trade-off confirmed; adder is not the main bottleneck (plain + gave 7.03 ns)
Shallower shifter: one-hot decode, two AND-OR levels	[4] Das & Khatri	About 10% faster for about 4% more area (up to 13.8% for 7.3%)	opt2 vs v1: 7.85 to 6.34 ns (−19.3%) for +2.5% area	Strongest result; biggest single win
Decode before the flops; B pre-XORed for subtract	[3] Uniyal; [7] Nagendra	[3]: −23.48% delay, −2.76% power	v1: 9.66 to 7.85 ns (mixed with the zero-flag change)	Decode sat on the critical path; gain not separable
Early zero flags: per unit, carry-free ADD, shift-zero mask	[3] (as above)	Take the post-mux zero check off the critical path	Whole chain 9.66 to 6.34 ns (−34%); power +39% (0.398 to 0.553 mW)	Delay met, power went the opposite way from [3]
Clock tightening: same RTL, shorter clock, antenna repair tuned	None (own experiment)	n/a	Passes 6.5 ns (+0.087 ns, 9 corners); fails 6.0 ns (−0.421 ns)	Proven result is 6.5 ns, only 1.3% slack

Paper Influence: Design B (Area & Power)

Change in our design	Paper	Paper expects	We measured	Conclusion
Compact adder: one shared RCA for add and sub (B inverted, carry-in = sub)	[1], [7]	RCA uses about half the CLA's power at similar delay (4-bit)	Whole design: area 5,514 to 4,299 µm² (−22%); power 0.398 to 0.331 mW (−17%)	Compact adder lowers area and power; cost is a slower clock
Shared shifter: one right-shifter, input and output reversed for left	[6] Kim & Kim	Sharing hardware cuts area (38% fewer transistors there)	Included in the −22% area; no separate number	Likely the second main area saving
Operand isolation: A_alu = A_reg & arithmetic_enable	[8] Banerjee	Saves power, but isolation costs area and timing	Runs 17 and 25: no reliable power saving, extra area	Agrees with the paper's warning, not its benefit
Remove redundant gating: Run 20 drops post-adder masks	[10] Ramkumar & Kittur	Less area and power for a slight delay increase	Delay proxy 9.66 to 11.32 ns (+17%); area and power fell	Direction as expected; delay penalty larger than 'slight'

B's savings come from less hardware, not from operand isolation. B was never run at a tight clock, so 11.32 ns is an upper bound.

Paper Influence: Design C (Hybrid)

Change in our design	Paper	Paper expects	We measured	Conclusion
Split adder: 8-bit RCA low half + 8-bit carry-increment high half	[6], [10]	Less area than a full carry-select with moderate delay	Whole-design numbers only (below)	Lands between B and A as intended
Precomputed subtract operands (B XOR SUB before the flops)	[3] Uniyal & Niranjan	Design adder and select logic together	Whole-design numbers only	Same trade as A's decode work, at lower cost
Shared rotator and shared P/G logic	[4], [6]	Shallower shifter, fewer duplicated gates	Whole-design numbers only	Keeps area down while holding delay

Hybrid at 20 ns	Area	Power	Delay proxy
C vs baseline (Run 7)	−11.5%	+19.5%	−14%
C vs B	+13.5%	+43.9%	−26.6%
C vs A	−22.5%	−13.9%	+31.1%

C is a balanced middle design: about 16% less energy per cycle than A at their fastest clocks, but not an optimum.

How We Use the Papers

What we can say

- The paper supports this technique, and our result moved in the same direction.
- Speed costs area and power (A); compact hardware saves area and power but slows the clock (B); C sits between.

What we do not claim

- That we implemented a paper or reproduced its number: the structures are not identical.
- That the shifter alone gave the 34% delay gain: decode and zero-flag changes were made together. The clean split is opt2 vs v1 (−19.3%).

Known weak points

- A plain + beat our hand CLA, so Yosys/ABC already builds a good adder.
- A's power rose 39% while [3] reported lower power (900 vs 627 cells, 68 vs 54 flops; vectorless estimate).
- B was never run at a tight clock, so the three-way tight-clock comparison is incomplete.

Common ALU Architecture

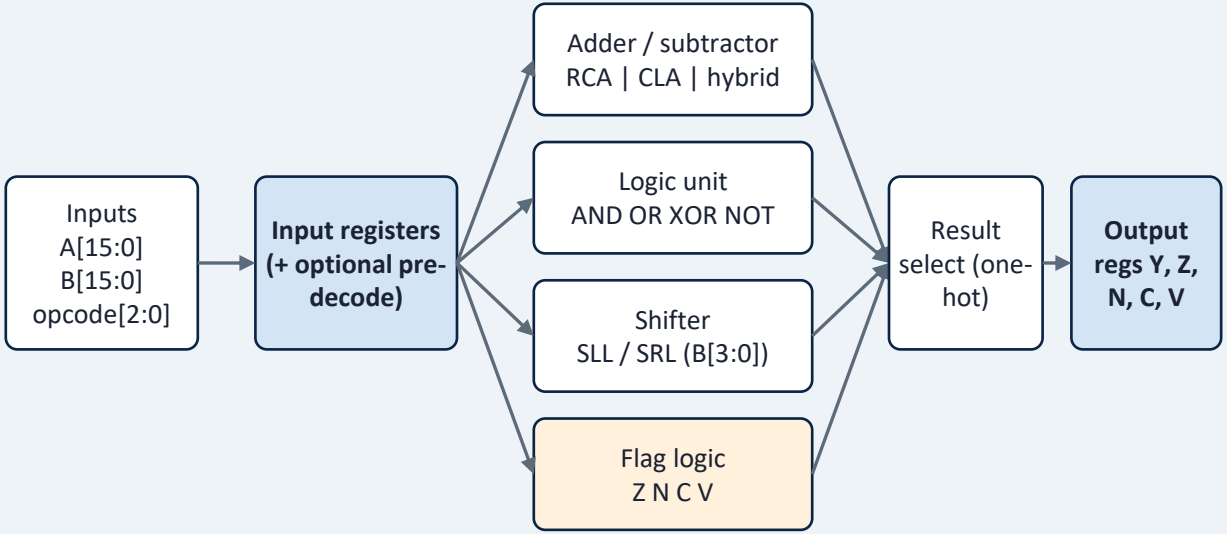


Figure 1: Common structure of the registered 16-bit ALU shared by all three designs

8

operations, 3-bit opcode

4

status flags: Z, N, C, V

2-stage

fixed latency, registered I/O

ALU Functional Specification

Opcode	Operation	Definition
000	ADD	A + B; carry and signed overflow reported
001	SUB	A − B (B inverted, carry-in = 1); C = no-borrow
010	AND	Bitwise A & B
011	OR	Bitwise A | B
100	XOR	Bitwise A ^ B
101	NOT A	Bitwise complement of A
110	SLL	Logical shift left by B[3:0]; C = last bit out
111	SRL	Logical shift right by B[3:0]; C = last bit out

Flags

Z result is zero

N result bit 15

C carry (no-borrow for SUB) or last bit shifted out

V signed overflow for ADD/SUB

Where We Started: The Baselines

Run	Adder	Area (µm²)	Setup slack (ns)	Est. period (ns)	Slew / fanout / antenna
Run 7	Behavioural + / −	5,514.04	+10.345	9.66	7 / 4 / 0
Run 8	Explicit 16-bit RCA	4,835.89	+9.678	10.32	0 / 4 / 0
Run 9	Four 4-bit-group CLA	5,336.37	+9.988	10.01	10 / 4 / 1

Table: the three simulation-verified baselines, all at 20 ns with 4 ns I/O delays (same flow for every later change)

What limited every baseline

Opcode decode, then the big output mux, then a 16-input zero check computed on the muxed result.

Phase 1 fixes (sub select, per-unit zero, CLA): WNS −3.18 to −0.41 ns.

Run 7: the reference point

5,514.04 µm², 0.3983 mW, 783 instances, 54 flops, setup +10.345 ns, hold +0.111 ns. Runs 1 to 6 were flow and setup clean-up.

RTL-to-GDSII Flow: Step by Step



1 Synthesis	2 Floorplan	3 Power grid	4 Placement	5 Clock tree (CTS)
Yosys + ABC turn RTL into a netlist of sky130_fd_sc_hd cells; ABC may restructure hand-written logic.	Sets die and core area from the target utilization (35%), places tap cells and I/O pins.	Builds the power distribution network (VDD/VSS rails and straps) over the core.	Global then detailed placement of cells (target density 40%), with timing-driven repair.	Builds the clock network for a real clock port; sink clustering 5, max diameter 20.

6 Routing	7 Parasitics + STA	8 GDSII stream-out	9 DRC	10 LVS + antenna
Global then detailed routing of signal nets, with antenna repair after global routing.	Extracts parasitics and runs static timing at nine corners (setup and hold).	Magic and KLayout write the final layout, alu16.gds.	Magic and KLayout design-rule checks must both report 0 violations.	Netgen compares layout with netlist (unique match required); antenna rules rechecked.

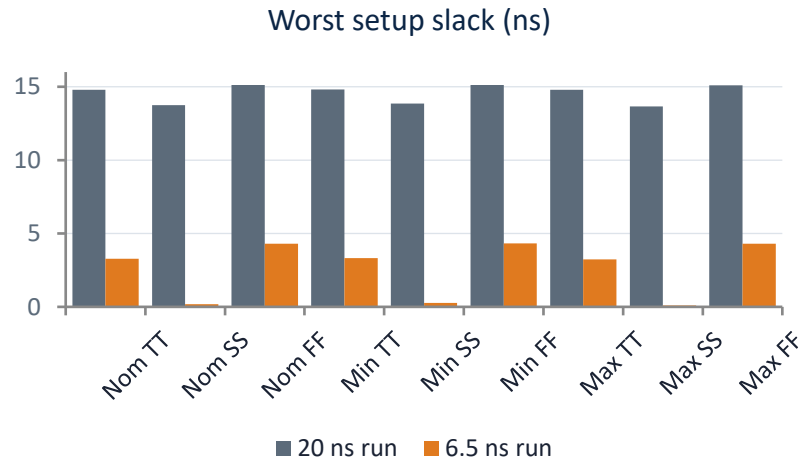
OpenLane 2 runs 74 steps in total on SKY130A; the ten stages above group them.

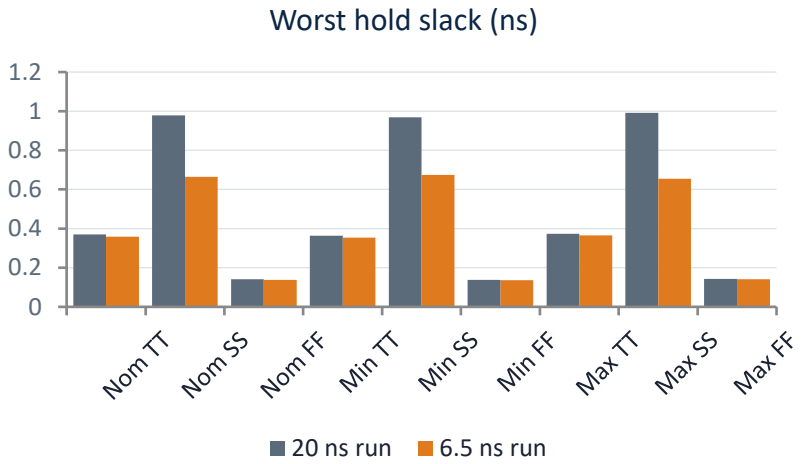
Run Configuration & Constraints

Setting	20 ns runs	Tight-clock runs (A 6.5 ns, C 8.5 ns)
Floorplan / placement	FP_CORE_UTIL 35, PL_TARGET_DENSITY_PCT 40	same
CTS	SINK_CLUSTERING_SIZE 5, MAX_DIAMETER 20	same
Antenna repair (global router)	defaults: margin 10, iterations 3	margin 40, iterations 10
Clock period (config and SDC agree)	20 ns	6.5 ns / 8.5 ns
Input / output delay	4 ns / 4 ns	1.3 ns / 1.3 ns (20% of 6.5 ns; 15% of 8.5 ns)
Uncertainty / transition / load	0.25 ns / 0.15 ns / 0.0334 pF	same
Max transition / capacitance / fanout	0.75 ns / 0.2 pF / 10	same

I/O delay was scaled with the clock: a fixed 4 ns made 6.0 ns look like a floor. CLOCK_PERIOD in config.json and create_clock in the SDC must agree.

STA Analysis: Nine Corners (Design A)





- **Limiting setup corner:** max_ss_100C_1v60 (slow, hot, low voltage): +0.087 ns at 6.5 ns, versus +3.28 ns typical and +4.31 ns fast.
- **Hold:** tightest at the fast corners (+0.136 to +0.143 ns); hold violations are repaired with delay cells.
- **Reading rule:** estimated period = clock − worst setup slack; only pass/fail at a given clock is solid.

Design A: Delay Progress



Checkpoints

- **Run 7:** 9.66 ns baseline.
- **v1:** decode before flops, 7.85 ns (−19%).
- **opt2:** one-hot shifters, 6.34 ns est. (−34%).
- **6.5 ns:** timing really met, +0.087 ns.

Design A: Finding the Bottleneck

Controlled experiment (10 ns)	RCA	CLA	Plain +
Isolated 16-bit adder, est. period	8.50 ns	7.16 ns	7.03 ns
Full ALU, est. period	9.99 ns	8.77 ns	8.67 ns

Table: CLA is about 16% shorter than RCA for the adder alone and about 12% for the full ALU, at 2 to 6% more area

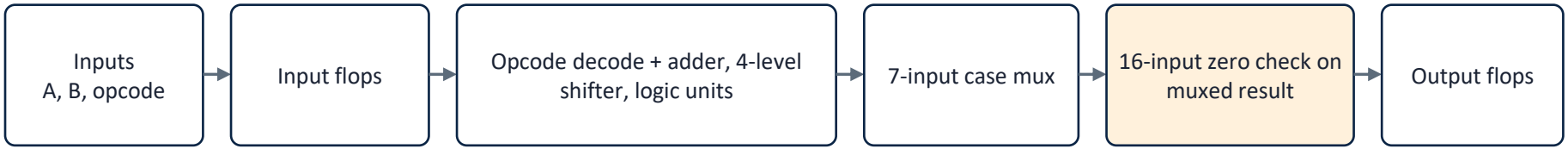
- **Phase 1:** every baseline hit the same path: opcode decode, big output mux, 16-input zero check on the muxed result.
- **Phase 2:** clocked ALU: early per-unit zero flags and a carry-free ADD zero check cut the period about 9%.
- **v1:** worst paths all began at the shift-amount flop and went through mux2, mux2, mux4; the CLA was about 1.5 ns behind.

The shifter, not the adder, was the bottleneck.

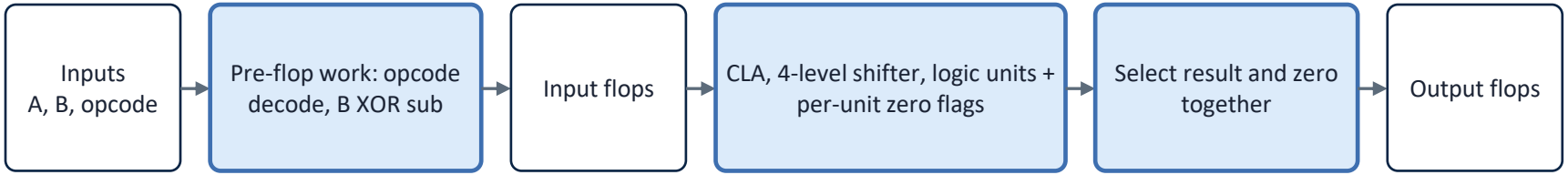
This finding decided opt2: attack only the shifter paths and the final select, keep everything else from v1.

Design A: Baseline to v1

Baseline (Run 7 / Run 9)



v1 (decode before the flops)



Result: 9.66 ns to 7.85 ns (−19%). Moving decode ahead of the flops is almost free because the input stage has a large time budget; the shifter now limits.

Design A: opt2, the Final Architecture

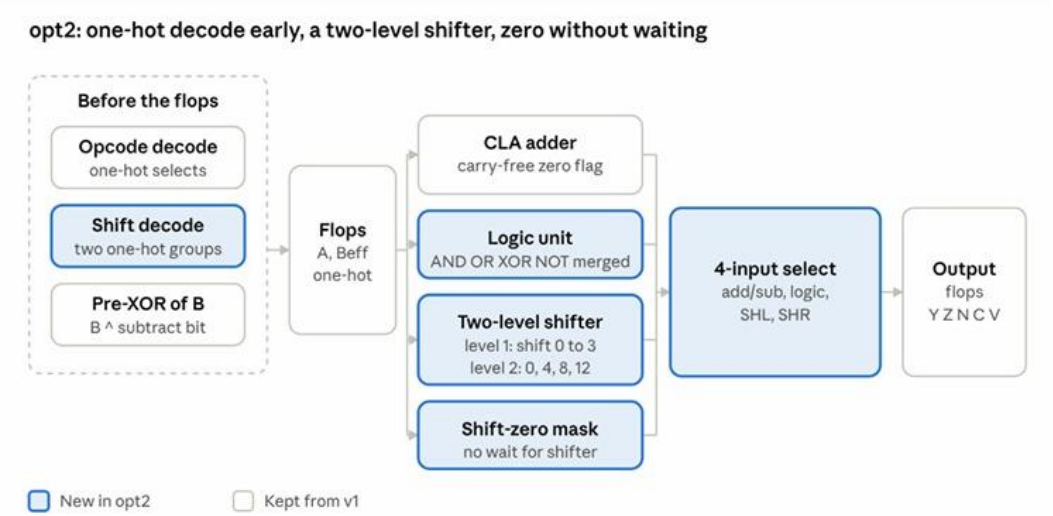


Figure: opt2 block diagram (blue = new in opt2, grey = kept from v1)

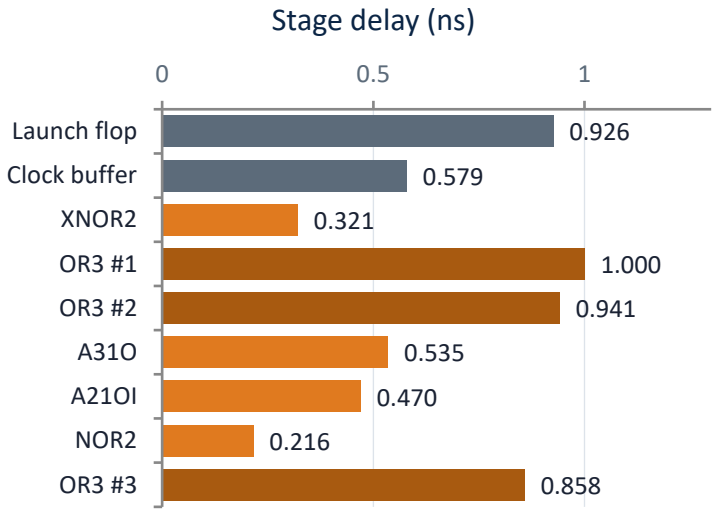
[4] Das & Khatri: merged, shallower shifter stages

Five changes from v1

- **One-hot shift decode** into two groups before the flops
- **Two-level AND-OR shifters** replace four mux levels [4]
- **Shift-zero from a mask,** no wait for the shifter
- **Shift carry-out** from the same selects
- **Merged logic unit:** 4-input final select

Design A: Delay Block by Block (6.5 ns)

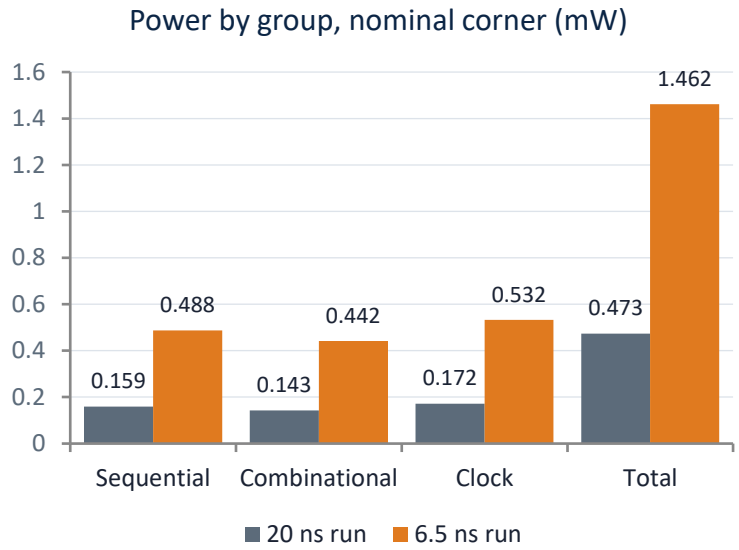
Instance	Cell	Fanout	Delay (ns)
_0955_ (launch flop, clk to Q)	dfxtp_1	5	0.926
fanout115	clkbuf_4	10	0.579
_0530_	xnor2_2	6	0.321
_0592_	or3_1	3	1.000
_0669_	or3_1	2	0.941
_0746_	a31o_1	3	0.535
_0747_	a21oi_2	5	0.470
_0749_	nor2_1	1	0.216
_0759_ (to result_comb[12])	or3_1	1	0.858



Worst setup path, max_ss_100C_1v60: path delay 5.846 ns, slack +0.087 ns. Launch flop and clock buffer take 1.505 ns (26%); logic 4.341 ns (74%). Three OR3 cells alone take 2.799 ns (48%).

Design A: Final Result & Power

Metric (opt2 @ 6.5 ns)	Final result
Setup timing	Met in all 9 corners, +0.087 ns
Hold timing	Met
Antenna	0 (margin 40, iterations 10)
DRC / LVS	0 / pass
Area	6,319.8 µm²
Cells / flops	902 / 68
Wirelength	21,550
Power	1.7085 mW (max corner), 1.4621 mW (nominal)
Max slew / fanout violations	21 / 1



Clock is 36% of total power and sequential cells 34% at both clocks; power scales with frequency.

Design A: Pushing to 6.0 ns (opt2 to opt5)

6.0 ns clock, slowest corner	opt2	opt3	opt4	opt5
Worst setup slack	−0.421 ns	−0.331 ns	−0.246 ns	−0.150 ns
Violating paths	11	37	6	9
Hold-fix buffers	0	40	9	4
Area (µm²)	6,317	7,420	6,092	6,357
Flops	68	82	83	102
Slew warnings	not recorded	17	8	0

- **opt3:** pre-registered adder terms and moved shifter and zero logic ahead of the flops; the input stage became the limit and 40 hold-fix buffers appeared.
- **opt4 / opt5:** rebalanced stages, then a shallow input stage with masks after the flops (102 flops).
- **Decision:** none passed 6.0 ns in a real run, so opt2 at 6.5 ns was frozen as the final design.

Design A: RTL Schematic

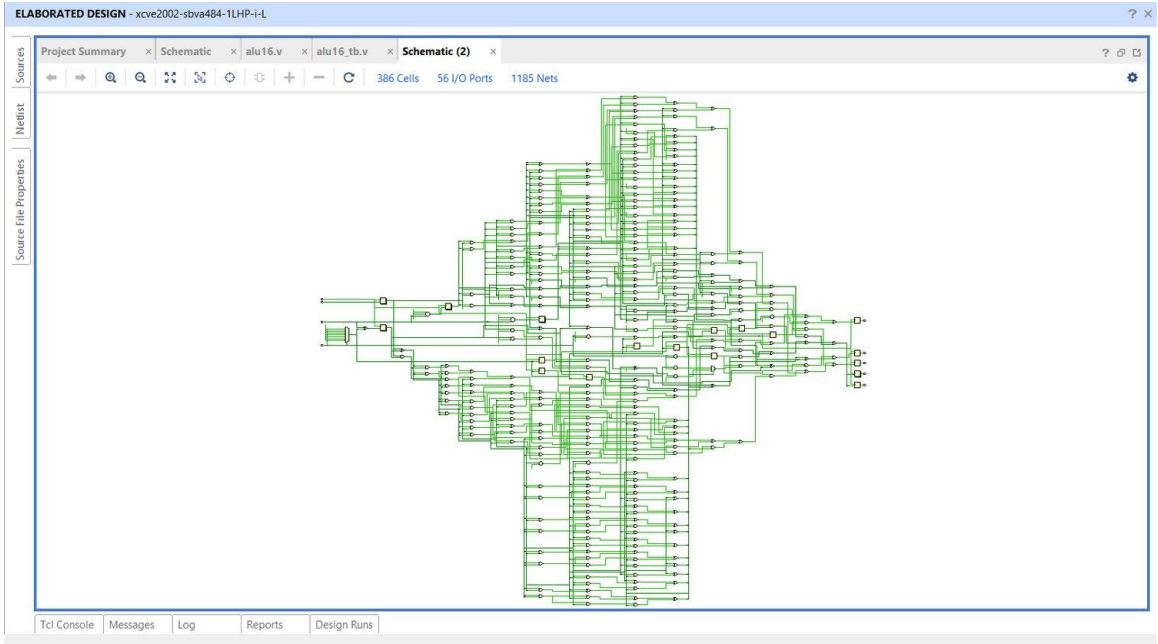


Figure: Vivado elaborated schematic of the delay design (delay_opt.v)

Elaborated RTL

- 386 cells, 1,185 nets
- Largest of the three designs (B 212, C 241 cells)
- Wide one-hot AND-OR shifter and per-unit zero logic are visible
- Verified: 13,375 test cases, ALL TESTS PASSED

Design B: Approach

- **Goal:** smallest area and lowest power at a fixed 20 ns clock; timing only needs to stay positive.
- **Start:** Run 7 behavioural baseline, 5,514.04 µm², 0.3983 mW (Runs 1 to 6 were flow clean-up).
- **Held constant:** sky130A / hd, utilization 35, density 40, CTS 5 / 20, same SDC for all 29 runs.
- **Step 1 (Runs 1 to 9):** adder choice, + vs RCA vs CLA.
- **Step 2 (Runs 10 to 19):** one-hot select, shared shifter.
- **Step 3 (Runs 20 to 29):** refinements; Run 20 selected.

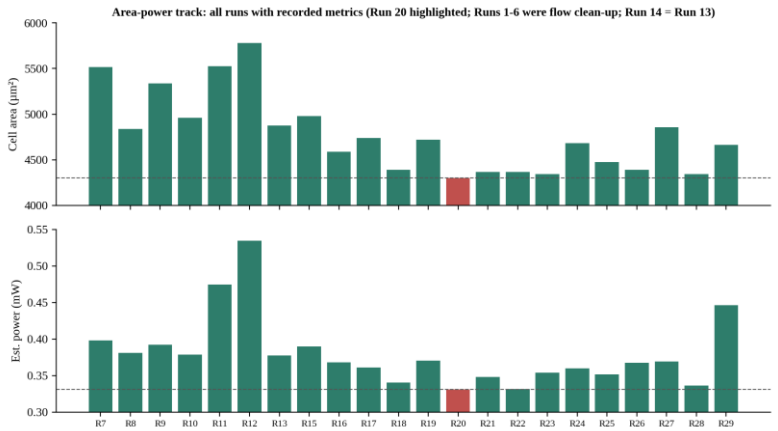


Fig: area and power per run; Run 20 (red) is selected, Run 14 equals Run 13.

−22.0%
area vs Run 7

−16.9%
power vs Run 7

Design B: Step 2 (Runs 10 to 19)

Run	Core idea	Area (µm²)	Power (mW)	Setup WNS	Decision
10	One-hot CLA / result selection	4,959.76	0.37835	+10.358	Useful checkpoint
11	Registered one-hot opcode predecode	5,527.80	0.47441	+11.599	Reject: register overhead
12	Registered conditional B operand	5,781.80	0.53449	+10.166	Reject: more state
13	CLA + shared bidirectional shifter	4,874.68	0.377213	+8.577	Retained
14	Balanced result network	= Run 13	= Run 13	= Run 13	No measured change
15	Partial opcode predecode	4,977.27	0.38982	+8.983	Reject
16	RCA + one-hot + shared shifter	4,588.15	0.36830	+7.173	Best at that point
17	Operand isolation	4,738.29	0.36073	+7.851	Power/area trade-off
18	Integrated shift carry	4,389.21	0.340132	+8.376	Best of phase
19	Selective B-register hold	4,718.28	0.37023	+7.240	Reject

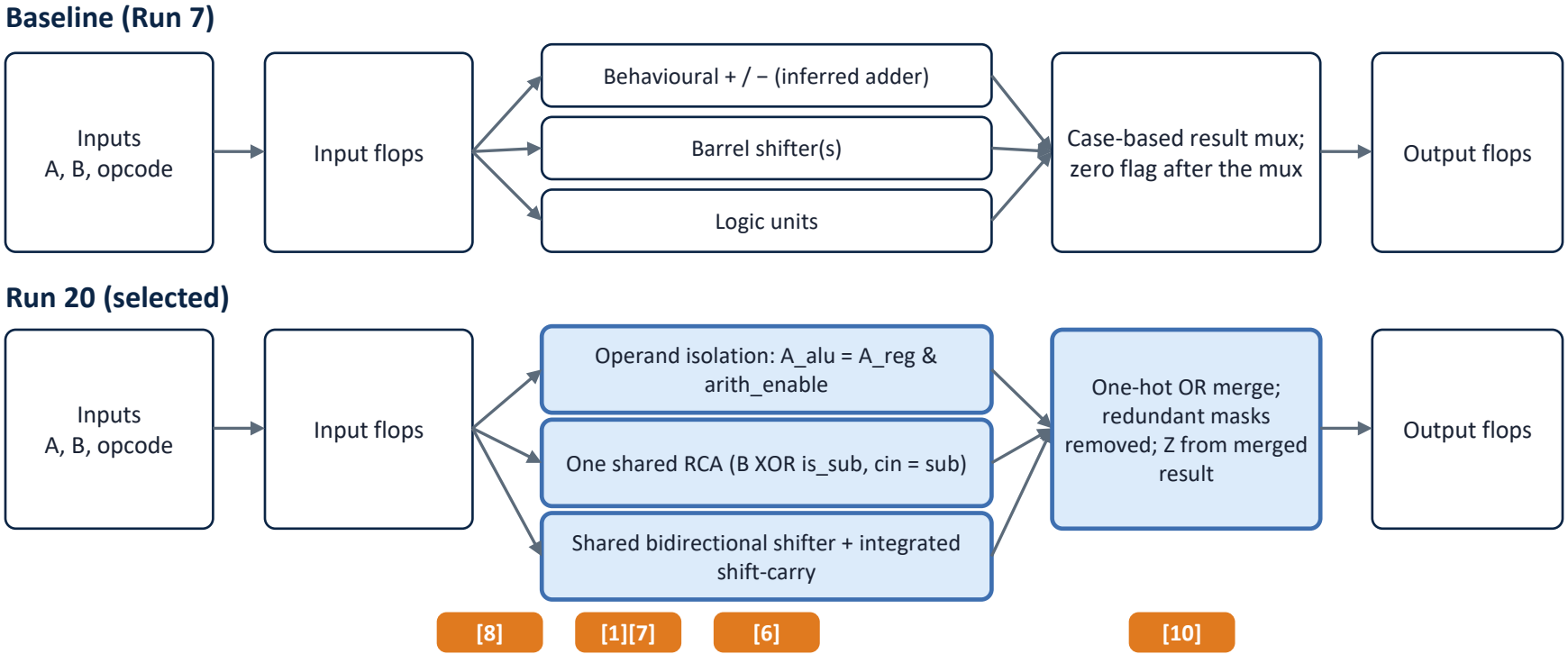
Registering decode or derived operands (Runs 11, 12) costs flops and clock load; sharing and integrating the shifter paid off (Runs 13, 18).

Design B: Step 3 (Runs 20 to 29)

Run	Hypothesis	Area (µm²)	Power (mW)	vs Run 20 (area / power)	Decision
20	Remove redundant arithmetic masks	4,299.12	0.330823	baseline	SELECTED
21	Case-based result selection	4,367.94	0.348211	+1.6% / +5.3%	Reject
22	Share Boolean intermediates	4,367.94	0.33156	+1.6% / +0.2%	Reject
23	Reuse RCA P/G for Boolean ops	4,345.42	0.354097	+1.1% / +7.0%	Reject
24	Inferred + adder	4,684.49	0.359632	+9.0% / +8.7%	Reject
25	Bitwise input isolation	4,473.04	0.351609	+4.1% / +6.3%	Reject
26	Flag factoring, no shift-carry mask	4,389.21	0.367542	+2.1% / +11.1%	Reject
27	Separate left/right shifters	4,855.91	0.368995	+13.0% / +11.5%	Reject
28	Reorder B isolation after XOR	4,344.17	0.336307	+1.0% / +1.7%	Reject (closest)
29	Registered one-hot opcode decode	4,663.22	0.446062	+8.5% / +34.8%	Reject (59 flops)

Why Run 20 works: with operand isolation and carry-in forced to zero, extra arithmetic masks are redundant; removing them cut area 2.05% and power 2.74% versus Run 18.

Design B: Baseline to Run 20

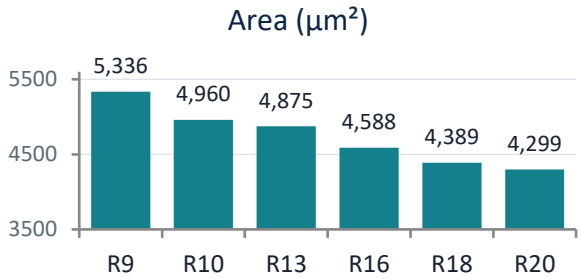


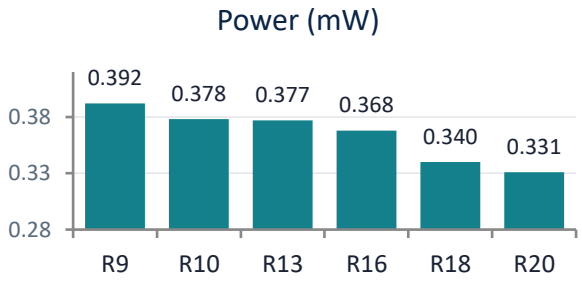
Blue = changed block; amber chips = supporting papers

Design B: Area & Power by Block Change

Block change	Runs	Δ Area (µm²)	Δ Power (mW)
Adder: inferred + to explicit RCA	7 to 8	−678.15 (−12.3%)	−0.0172 (−4.3%)
One-hot result select	9 to 10	−376.61 (−7.1%)	−0.0141 (−3.6%)
Shared bidirectional shifter	10 to 13	−85.08 (−1.7%)	−0.0011 (−0.3%)
CLA replaced by RCA	13 to 16	−286.53 (−5.9%)	−0.0089 (−2.4%)
Integrated shift carry	16 to 18	−198.94 (−4.3%)	−0.0282 (−7.6%)
Redundant masks removed	18 to 20	−90.09 (−2.1%)	−0.0093 (−2.7%)

Run-to-run deltas from the measured sequence; other source changes interact, so this is not an exact per-block split. Run 20: 627 instances, 54 flops, 4,299.12 µm², 0.3308 mW.





Total vs Run 7: −1,214.92 µm² (−22.0%), −0.0675 mW (−16.9%)

Design B: Run 20 Final Numbers

Metric	Run 20 (20 ns)
Cell area / instances	4,299.12 µm² / 627 (54 flops)
Estimated power	0.330823 mW (nominal 0.2834 mW)
Setup / hold WNS	+8.676 ns / +0.142 ns
Functional regression	13,375 cases passed; 150,000-cycle stress: 0 mismatches
Max slew / fanout / cap violations	0 / 4 / 0
Antenna / DRC / Magic / KLayout / LVS	0 / 0 / 0 / 0 / pass
Flow errors	0 (one warning)

Final RTL: area_opt_alu16_rca_run20.v

−22.0 %
area vs Run 7

−16.9 %
power vs Run 7

Honest limits

- 4 max-fanout violations remain.
- Z is still taken after the merge: the main reason B is slow.
- Only ever run at 20 ns (+8.7 ns spare); tight-clock timing not measured.

Design B: RTL Schematic

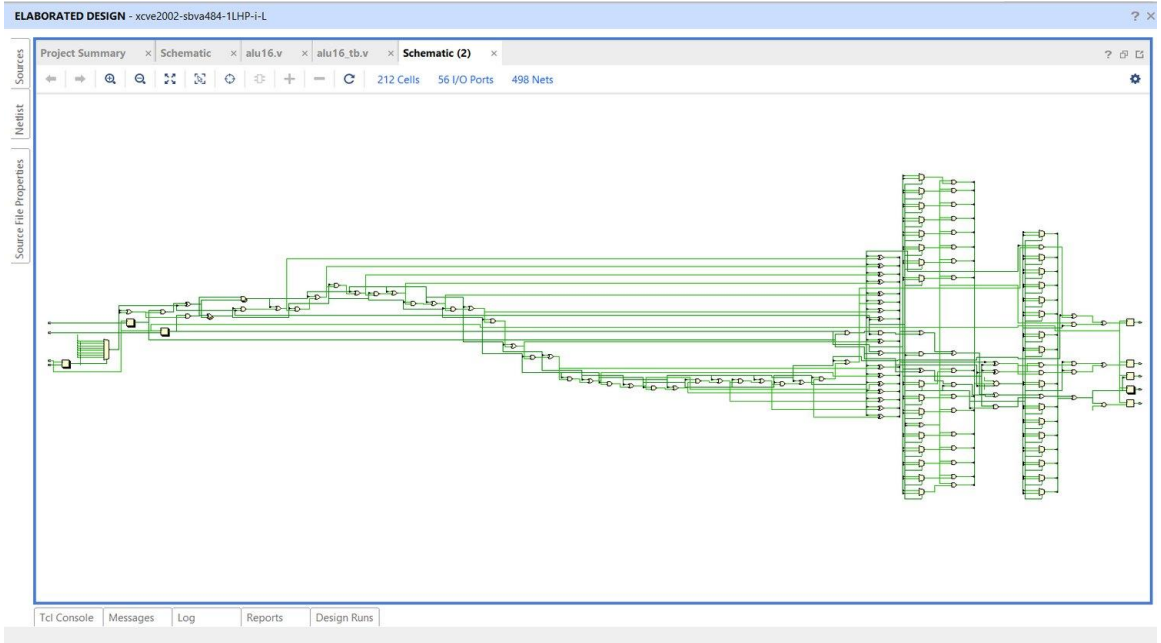


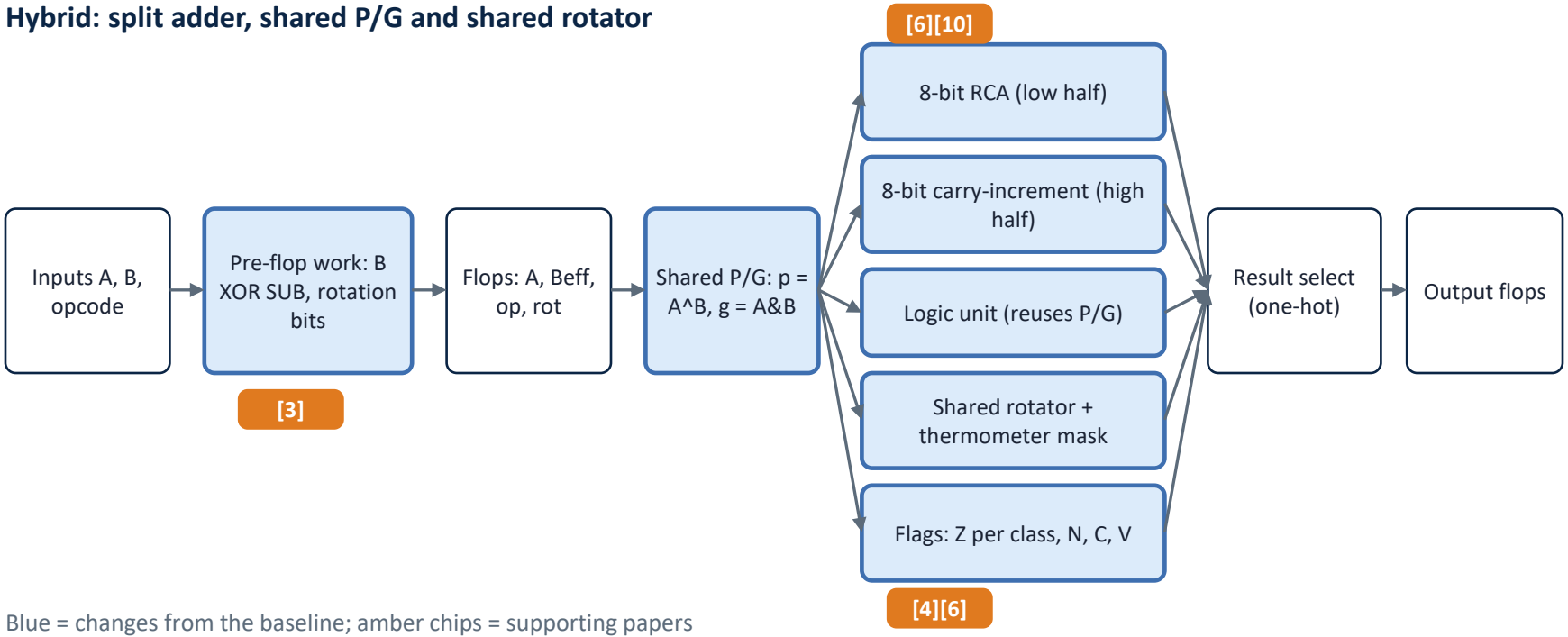
Figure: Vivado elaborated schematic of Run 20 (area-power design)

Elaborated RTL

- 212 cells, 498 nets
- Smallest of the three designs
- One shared adder and one shared shifter block
- Verified: 13,375 test cases, ALL TESTS PASSED

Design C: Hybrid Architecture

Hybrid: split adder, shared P/G and shared rotator



Blue = changes from the baseline; amber chips = supporting papers

Design C: RTL Schematic

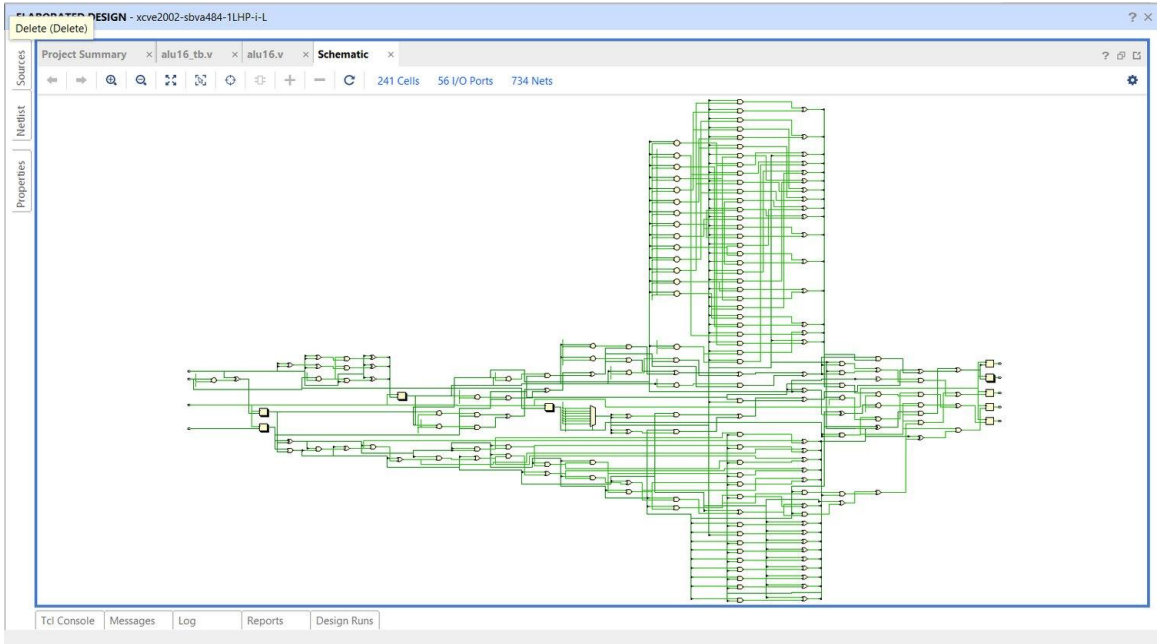


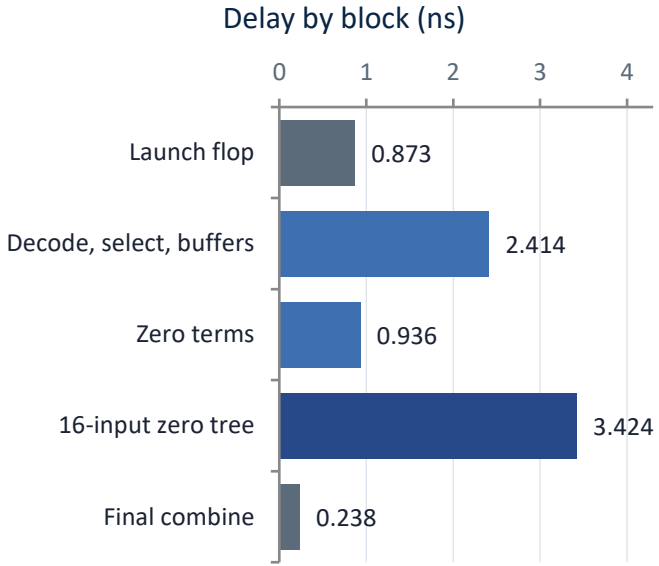
Figure: Vivado elaborated schematic of the hybrid design (hybrid_opt.v)

Elaborated RTL

- 241 cells, 734 nets
- Between A (386) and B (212)
- Split adder and shared rotator visible in the datapath
- Verified: 13,375 test cases, ALL TESTS PASSED

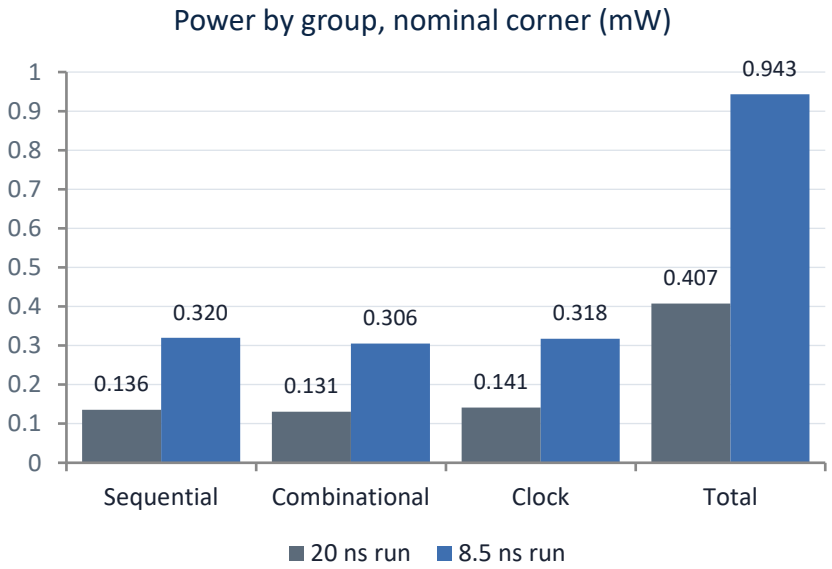
Design C: Delay Block by Block (8.5 ns)

Stage (max_ss_100C_1v60)	Cell	Delay (ns)
Clock to Q of op_reg[2]	dfxtp_1	0.873
Opcode decode	and2b_1	0.572
Fanout buffer	clkbuf_4	0.447
Select logic	a21boi_4	0.565
Fanout buffer	clkbuf_4	0.595
Inverter	inv_2	0.235
Shift / logic zero terms	a2bb2o_1 + a31o_1	0.557 + 0.379
Zero tree, level 1	or4_1	1.304
Zero tree, level 2	or4_1	1.160
Zero tree, level 3	or3_1	0.960
Final combine into zero_comb	o21ai_1	0.238



Slack +0.166 ns, arrival 8.482 ns. The three OR cells of the zero tree take 3.42 ns, about 40% of the path; the adder is not the limiter. The listed stages sum to 7.885 ns.

Design C: Power & Area by Block



Area and cells	8.5 ns	20 ns
Cell area (µm²)	4,863.41	4,878.43
Core / die (µm²)	13,190 / 17,378	13,190 / 17,378
Utilization	36.87%	36.99%
Instances	685	686
Timing-repair buffers	81	81
Clock buffers / inverters	11 / 4	13 / 3
Routed wirelength (µm)	14,820	15,307

Clock network is about 34% of power and sequential about 34% in both runs. Max-corner totals: 1.1016 mW at 8.5 ns, 0.4761 mW at 20 ns.

Design C: Timing & Sign-off Numbers

Timing metric	8.5 ns	20 ns
Worst setup slack	+0.1657 ns	+11.6927 ns
Worst hold slack	+0.1490 ns	+0.1542 ns
Setup / hold violations	0 / 0	0 / 0
Worst setup corner	max_ss	max_ss
Est. minimum period	8.3343 ns	8.3073 ns

Check	8.5 ns	20 ns
Routing DRC	0	0
Magic / KLayout DRC	0 / 0	0 / 0
LVS	PASS	PASS
Antenna / pins / grid	0 / 0 / 0	0 / 0 / 0
Max cap / fanout	0 / 0	0 / 0
Max transition	3 (max_ss)	0

Both runs share a byte-identical Yosys netlist, so the comparison isolates the clock. The 8.5 ns run's 3 max-transition violations are about 0.7528 ns against a 0.750 ns limit: setup/hold closed, but not all electrical checks clean.

Architecture Comparison

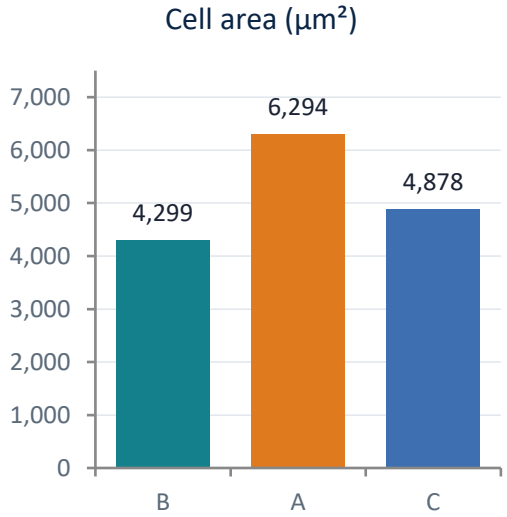
Feature	A: Delay	B: Area + power	C: Hybrid
Adder	Grouped CLA (4 x 4-bit)	16-bit RCA	8-bit RCA + 8-bit carry-increment
Shifter	Two one-hot AND-OR shifters, 2 levels	One shared bidirectional shifter	One shared rotator + thermometer mask
Operand isolation	No	Yes	No (removed)
Logic unit	Merged AND/OR/XOR/NOT	Bitwise paths into one-hot select	Reuses adder P/G
Zero flag	Per unit, shift zero from mask	Per result path (after merge)	Per class, operand/mask based
Pre-register work	Opcode classes, B XOR SUB, one-hot shift decode	Decode after flops	B XOR SUB, shift rotation bits

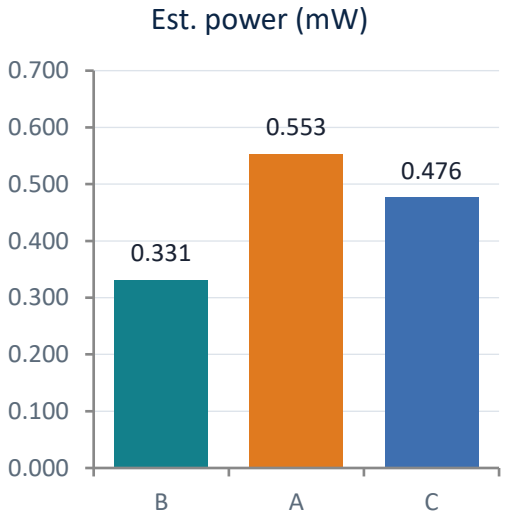
Results: Three-Way Comparison

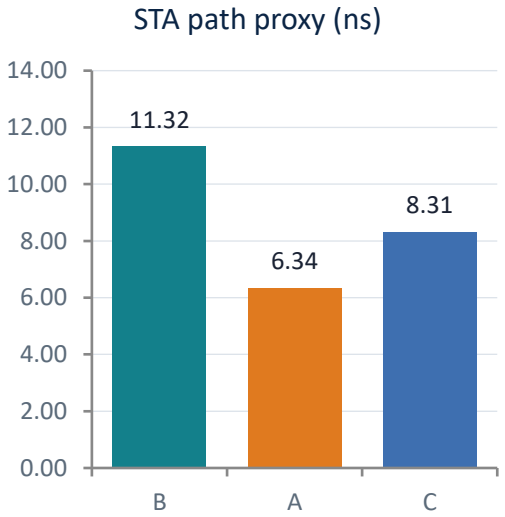
Metric (20 ns clock)	B: Area + power	A: Delay	C: Hybrid
Cell area (µm²)	4,299.12	6,293.54	4,878.43
Instances / flops	627 / 54	900 / 68	686 / 57
Estimated power (mW)	0.3308	0.5532	0.4761
Worst setup slack (ns)	+8.676	+13.663	+11.693
STA path proxy (ns)	11.32	6.34	8.31
Max-slew / max-fanout	0 / 4	31 / 1	0 / 0
DRC / LVS errors	0 / 0	0 / 0	0 / 0
Area × proxy (derived, µm²·ns)	about 48,700	about 39,900	about 40,500

STA path proxy = clock period − worst setup slack; power from the max-power corner (vectorless estimate)

Results: Area, Power and Timing at 20 ns



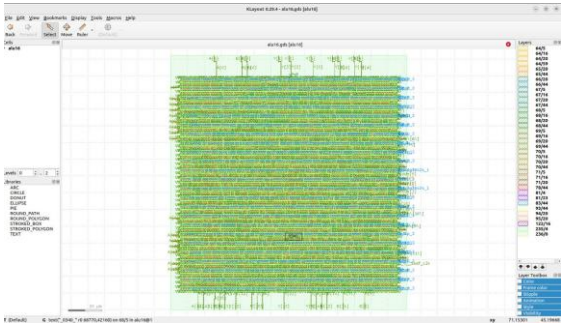




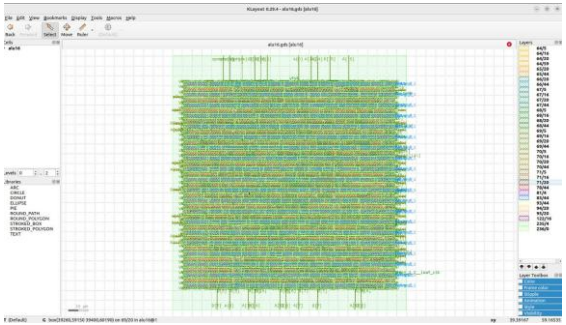
B Area + power **A** Delay **C** Hybrid

B is smallest and lowest-power but slowest; A is fastest but largest and highest-power; C sits between.

Physical Implementation (GDSII)



Design A (6.5 ns)



Design B (20 ns)



Design C (8.5 ns)

Design	Clock	Area (µm²)	Setup slack	DRC / LVS / antenna
A: Delay	6.5 ns	6,319.81	+0.087 ns	Passed
B: Area + power	20 ns	4,299.12	+8.676 ns	Passed
C: Hybrid	8.5 ns	4,863.41	+0.166 ns	Passed

KLayout views are zoom-to-fit; die size is not comparable between views

Closure Demonstrated & Energy per Cycle

Design	Shortest clock passed	Margin	Area (μm²)	Power, nominal (max corner)
A: Delay	6.5 ns (6.0 ns failed by 0.421 ns)	+0.0873 ns	6,319.8	1.4621 mW (1.7085)
C: Hybrid	8.5 ns	+0.1657 ns	4,863.41	0.9429 mW (1.1016)
B: Area-power	20 ns only (never tightened)	+8.676 ns	4,299.12	0.2834 mW (0.3308)

Energy view	B	A	C	Lowest
Energy per cycle at 20 ns (pJ)	5.67	9.47	8.15	B
Power × STA proxy (mW·ns)	3.21	3.00	3.38	A
Energy at each design's fastest clock (pJ)	5.67 (20 ns)	9.50 (6.5 ns)	8.01 (8.5 ns)	B; C below A

A and C ran with identical constraints (1.3 ns I/O, CTS 5/20, antenna 40/10); B's tight-clock result is not measured.

Results: Key Observations

No dominant design

Area, power and delay cannot all be won at once.

C vs B

26.6% shorter proxy for +13.5% area and +43.9% power.

C vs A

22.5% less area and 13.9% less power, 31.1% longer proxy.

Closure shown

A at 6.5 ns, C at 8.5 ns; B was run only at 20 ns.

Energy per cycle

At each fastest closed clock, C is about 16% below A (8.01 vs 9.50 pJ).

Common limiter

Once adder and shifter are fixed, opcode decode and zero-flag logic set the clock.

Real-World Applications of the Three Designs

A: Faster logic

Typical uses

- High-throughput DSP pipelines
- Network packet processing
- High-frequency processor execution units
- Real-time control loops

Why it fits

Shortest clock: closes at 6.5 ns (about 154 MHz on SKY130A). Cost: largest area, 6,294 µm², and 0.553 mW at 20 ns.

B: Compact, low power

Typical uses

- Battery-powered IoT sensor nodes
- Wearables and medical or hearing devices
- RFID tags and smart cards
- Always-on, low-duty-cycle microcontrollers

Why it fits

Smallest and lowest-power: 4,299 µm² and 0.331 mW at 20 ns. Only verified at 20 ns, so suited to relaxed clocks.

C: Balanced

Typical uses

- General-purpose embedded microcontrollers
- Motor control and industrial controllers
- Audio and sensor-hub processors
- Edge devices and teaching cores

Why it fits

Middle on every axis: closes at 8.5 ns (117.65 MHz), 4,878 µm², 0.476 mW; about 16% less energy per cycle than A at their fastest clocks.

Illustrative use cases drawn from the measured trade-offs; the designs were not tested in these applications. All numbers are SKY130A tool estimates.

Conclusion & Future Scope

Conclusion

- One 16-bit, 8-operation ALU taken from RTL to DRC/LVS-clean GDSII in three variants; none dominates every metric.
- A met 6.5 ns; B is 22% smaller than the baseline; C is balanced and met 8.5 ns.
- Each change was guided by a paper and measured; results move in the papers' direction.
- All numbers are tool estimates, not silicon measurements.

Future Scope

- Run Design B at the tight clocks used for A and C.
- Clear the remaining slew, fanout and transition violations.
- Restructure opcode decode and zero-flag logic.

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Thank You

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